

Sampling ~~Chruque~~

Register - 02

Disc. Stats Part 2

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Roll # 23

Probability
Sampling

Simple Random Sampling:

Def: Simple random sampling is the simple and basic technique of probability sampling where each and every sampling unit of the population has the equal/same chance or probability of being selected for the sample and every sample from all the possible samples has the same chance of being selected, then the sample is known as "simple random sample".

N: Sampling unit in the population

P_i = Prob. of "Sampling unit being selected in sample" = $\frac{1}{N}$

In case of without replacement:-

All possible sample - APS = $\binom{N}{n}$

$$P_{\text{sample}} = P_i = \frac{1}{\binom{N}{n}}$$

In case of with replacement:-

All possible sample - APS = N^n

$$P_{\text{sample}} = P_i = \frac{1}{N^n}$$

Sampling with replacement the population does not change trial, therefore, the

Finite Population Correction Factor:

For a random sample of size 'n' from an infinite population, it is well known that the variance of the mean is σ^2/n . The only change in this result when the population is finite is the introduction of the factor $(N-n)/N$. The factors $(N-n)/N$ for the variance and $\sqrt{(N-n)/N}$ for the standard error are called the finite population ^{correction} factor (fpc).

Provided that the sampling fraction n/N remains low, these factors are close to unity, and the size of the population as such has no direct effect on the standard error of the sample mean. For instance, if S is the same in the two populations, a sample of 500 from a population of 200,000 gives almost as precise an estimate of the population mean as a sample of 500 from a given population of 10,000.

In practice the fpc can be ignored whenever the sampling fraction does not exceed 5% (means fpc ignored when $f \leq 0.05$).

The effect of ignoring the correction is to overestimate the standard error of the estimate $\bar{y}$. (mean when we ignore fpc then variance becomes little bit high).

$$\left(\frac{N-n}{N} \right) = 1 - f = \left(1 - \frac{n}{N} \right)$$

$$\sigma_{\bar{y}}^2 = \frac{\sigma^2}{n}$$

Estimation Of Variance:

True Variance = $V(\bar{y})_{\text{true}} = \left(\frac{N-n}{N \cdot n} \right) S^2$ where $S^2 = \frac{\sum (y_i - \bar{y})^2}{N-1}$

An estimated variance is:

$$V(\bar{y}) = \left(\frac{N-n}{N \cdot n} \right) \text{Est. } S^2 \quad \text{Est. = Estimator}$$

$$V(\bar{y}) = \left(\frac{N-n}{N \cdot n} \right) S^2 \quad \text{where } S^2 = \frac{\sum (y_i - \bar{y})^2}{n-1}$$

take expectation $E[V(\bar{y})] = \left(\frac{N-n}{N \cdot n} \right) E(S^2) \therefore E[V(\bar{y})] = \left(\frac{N-n}{N \cdot n} \right) S^2$

$$E[V(\bar{y})] = V(\bar{y})$$

$$\text{Est } S^2 = S^2$$

$$E(S^2) = S^2$$

Theorem: Prove that the sample variance S^2 is an unbiased estimator of the population variance S^2 .

Proof: we define

$$S^2 = \sum_{i=1}^n (y_i - \bar{y})^2 \quad \text{--- (I)}$$

consider $\sum (y_i - \bar{y})^2$ and add & subtract $\bar{y}$.

$$\sum (y_i - \bar{y})^2 = \sum (y_i - \bar{y} + \bar{y} - \bar{y})^2$$

$$= \sum [(y_i - \bar{y}) - (\bar{y} - \bar{y})]^2$$

$$= \sum [(y_i - \bar{y})^2 + (\bar{y} - \bar{y})^2 - 2(y_i - \bar{y})(\bar{y} - \bar{y})]$$

The Effect Of Change in the Value of P on standard Error:

P: Values of the True %age

P	10	20	30	40	50	60	70	80	90
Q	90	80	70	60	50	40	30	20	10
Var = PQ	900	1600	2100	2400	2500	2400	2100	1600	900
SE(P) = $\sqrt{\frac{PQ}{n}}$	30	40	46	49	50	49	46	40	30

$$S.E(P) = \sqrt{\frac{PQ}{n}}$$

For fixed Sample size $n=100$

$$S.E(P) = \sqrt{PQ}$$

- 1: Standard error is maximum at point $P = 50\%$
- 2: Change in the standard error is symmetrical about the point $P = 50\%$
- 3: If true proportion P lies between 30% and 70%, the standard error is not effected greatly.

This approach is not appropriate when interest lies in the total number of units in the population that fall in class 'C'. In this event we think of the S.E expressed as the percentage of the true value NP. This

we will talk about co-efficient of variation of the estimate $\hat{A}$.

Now
$$C.V(\hat{A}) = \frac{S.E(\hat{A})}{\hat{A}} = \frac{S.E(\hat{A})}{NP} \quad \because \hat{A} = NP$$

$$= \frac{N \left[\frac{N-n}{N-1} \right] \sqrt{\frac{PQ}{n}}}{NP} \quad \therefore V(\hat{A}) = N \left[\frac{N-n}{N-1} \right] \frac{PQ}{n}$$

$$S.E(\hat{A}) = N \left[\frac{N-n}{N-1} \right] \sqrt{\frac{PQ}{n}}$$

For large N ; $N-1 = N$

$$C.V(\hat{A}) = \sqrt{\frac{N-n}{N}} \sqrt{\frac{PQ}{n}}$$

Ignoring f.p.c

$$C.V(\hat{A}) = \sqrt{\frac{PQ}{n}}$$

$$C.V(\hat{A}) = \sqrt{\frac{Q}{nP}}$$

For Sample Size $n=1$

$$C.V(\hat{A}) = \sqrt{\frac{Q}{P}}$$

Values of P in Percentage

P	0.1	0.5	1.0	5.0	10.0	20.0	30.0	40.0	50	60	70
Q	99.9	99.5	99	95	90	80	70	60	50	40	30
$\sqrt{Q/P}$	31.6	14.1	9.9	4.4	3.0	2.0	1.5	1.2	1.0	0.8	0.7

The co-efficient is high when P is less than 5%.

Very large samples are needed for precise estimates of the total number possessing any attribute i.e. rare in the population [say (A)].

Simple random sampling or any method of sampling that is employed for general purpose, is an expensive method of estimating the total number of units of rare type. Therefore some alternative method is used in such situations.

~~Theorem~~ Show that the covariance of independent sample (W.R.) is equal to zero i.e. or $\text{Cov}(y_i, y_j) = 0$

Proof:

$$\text{Cov}(y_i, y_j) = E[(y_i - \bar{y})(y_j - \bar{y})]$$

$$= E[y_i y_j - y_i \bar{y} - \bar{y} y_j + \bar{y}^2]$$

$$= E[y_i y_j] - E[y_i \bar{y}] - E[\bar{y} y_j] + E[\bar{y}^2]$$

As $\bar{y}$ is a constant

$$= E[y_i y_j] - E[y_i] \bar{y} - E[\bar{y} y_j] + E[\bar{y}^2]$$

$$= \frac{1}{N} \sum y_i y_j - \frac{1}{N} \sum y_i \bar{y} - \frac{1}{N} \sum \bar{y} y_j + \frac{1}{N} \sum \bar{y}^2$$

$$= \frac{1}{N} \sum y_i y_j - \frac{1}{N} \sum y_i \bar{y} - \frac{1}{N} \sum \bar{y} y_j + \frac{1}{N} \sum \bar{y}^2$$

$$= \frac{1}{N} \sum y_i y_j - \frac{1}{N} \sum y_i \bar{y} - \frac{1}{N} \sum \bar{y} y_j + \frac{1}{N} \sum \bar{y}^2$$

$$= \frac{1}{N} \sum y_i y_j - \frac{1}{N} \sum y_i \bar{y} - \frac{1}{N} \sum \bar{y} y_j + \frac{1}{N} \sum \bar{y}^2$$

$$= \frac{1}{N} \sum y_i y_j - \frac{1}{N} \sum y_i \bar{y} - \frac{1}{N} \sum \bar{y} y_j + \frac{1}{N} \sum \bar{y}^2$$

S.E is due to one fact and that is our survey based on sample and S.E measure by formula

i.e; statistic - Parameter

How P-value effect sample size? (i) bill)

Estimation of Sample Size

Too large a sample is wastage of resources

e.g:- $n = 10,000$

$$S.E(p) = 0.15$$

$$n = 5000$$

$$S.E(p) = 0.19$$

There is small difference in S.E in both cases so it is wise decision to take small sample size by compromising the small increase in S.E. No doubt with very large sample more precise result is obtained but to handle very large sample is difficult. For large sample more time and money is required.

Too small a sample may produce the result with low precision

e.g:

$$n = 200$$

$$S.E(p) = 21.5$$

We intend to maintain a balance between the cost/resources and precision

So, some moderate value for sample size should be decided for sample survey. This decision varies from one situation to other. So we conclude that both extremes are not wise. So we should decide moderate sample size for sample survey.

6: Finally the chosen value of 'n' must be appraised/evaluate to see whether it is consistent with the resources available to take the sample. This demands an estimation of the cost, labour, time and materials required to obtain the proposed size of sample. It sometimes becomes apparent that 'n' will have to be drastically reduced. (A hard decision must then be faced, whether to proceed with a much smaller sample size, thus reducing precision, or to leave/abandon efforts until more resources can be found.)

Derivation Of Formula For estimation of 'n' with continuous data: ^{20A Q2(b)}

Case-I: If we wish to control the margin of error 'd' in the estimated population mean. With simple random sampling having sample mean $\bar{y}$, we want

$$Pr(|\bar{y} - \bar{Y}| > d) = \alpha \text{ where } \alpha \text{ is a small probability}$$

we assume that $\bar{y}$ is normally distributed and its

$$V(\bar{y}) = \frac{(N-n)S^2}{N \cdot n}$$

The eq. that connects 'n' with the desired level of precision is;

$$Z = \frac{|\bar{y} - \bar{Y}|}{S.E(\bar{y})} = \frac{d}{\sqrt{\frac{(N-n)S^2}{N \cdot n}}}$$

Squaring both sides we get

$$Z^2 = \frac{d^2}{\left(\frac{N-n}{N}\right) \frac{S^2}{n}}$$

$$Z^2 = \frac{d^2}{\frac{(N-n)S^2}{Nn}}$$

$$\frac{Z^2 S^2}{d^2} = \frac{1}{\frac{(N-n)}{Nn}}$$

$$\frac{Z^2 S^2}{d^2} = \frac{Nn}{N-n}$$

$$\frac{1}{N} \left(\frac{Z^2 S^2}{d^2} \right) = \frac{n}{N-n}$$

$$\frac{1}{\frac{1}{N} \left(\frac{Z^2 S^2}{d^2} \right)} = \frac{N-n}{n}$$

$$\frac{1}{\frac{1}{N} \left(\frac{Z^2 S^2}{d^2} \right)} = \frac{N-n}{n}$$

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$$\frac{1}{\frac{1}{N} \left(\frac{Z^2 S^2}{d^2} \right)} = \frac{N-n}{n}$$

$$1 + \frac{1}{\frac{1}{N} \left(\frac{Z^2 S^2}{d^2} \right)} = \frac{N}{n}$$

$$\frac{1}{\frac{1}{N} \left(\frac{Z^2 S^2}{d^2} \right)} = \frac{n}{N}$$

$$1 + \frac{1}{\frac{1}{N} \left(\frac{Z^2 S^2}{d^2} \right)} = \frac{N}{n}$$

$$\frac{\frac{N}{n} \left(\frac{Z^2 S^2}{d^2} \right)}{1 + \frac{1}{\frac{1}{N} \left(\frac{Z^2 S^2}{d^2} \right)}} = n$$

$$\Rightarrow n = \frac{Z^2 S^2 / d^2}{1 + \frac{1}{\frac{1}{N} \left(\frac{Z^2 S^2}{d^2} \right)}}$$

Case II: (for relative error 'r'): Q2(b)

If we wish to control the relative error 'r' in the estimated population mean. With simple random sampling having mean $\bar{y}$, we want

$$P_r(|\bar{y} - \bar{Y}| > r\bar{Y}) = \alpha$$

where α is a small probability.

we assume that $\bar{y}$ is normally distributed and its

$$V(\bar{y}) = \frac{(N-n)S^2}{Nn}$$

The eq. that connects 'n' with the desired level of precision

$$z = \frac{|\bar{y} - \bar{Y}|}{S.E(\bar{y})}$$

$$z = \frac{r\bar{Y}}{\sqrt{\frac{(N-n)S^2}{Nn}}}$$

Ignoring F.P.C

$$n_0 = \frac{z^2 S^2}{d^2}$$

$$n_1 = \frac{n_0}{1 + \frac{n_0}{N}} \quad \text{if } n_1 < n_0$$

Sq. both sides, we get

$$z^2 = \frac{r^2 \bar{Y}^2}{\frac{(N-n)S^2}{Nn}}$$

$$n = \frac{n_1}{1 + \frac{n_1}{N}} \rightarrow n_2 < n_1$$

$$\frac{z^2 S^2}{r^2 \bar{Y}^2} = \frac{nN}{N-n}$$

$$\frac{1}{N} \left(\frac{z^2 S^2}{r^2 \bar{Y}^2} \right) = \frac{N-n}{n}$$

$$\frac{1}{N} \left(\frac{z^2 S^2}{r^2 \bar{Y}^2} \right) = \frac{N-1}{n}$$

$$\frac{1}{N} \left(\frac{z^2 S^2}{r^2 \bar{Y}^2} \right) + 1 = N$$

$$1 + \frac{1}{N} \left(\frac{z^2 S^2}{r^2 \bar{Y}^2} \right) = \frac{N}{n}$$

$$\frac{1}{N} \left(\frac{z^2 S^2}{r^2 \bar{Y}^2} \right) = \frac{N}{n} \Rightarrow n = \frac{z^2 S^2}{r^2 \bar{Y}^2}$$

$$1 + \frac{1}{N} \left(\frac{z^2 S^2}{r^2 \bar{Y}^2} \right) = \frac{N}{n} \Rightarrow n = \frac{z^2 S^2}{r^2 \bar{Y}^2 \left(1 + \frac{1}{N} \left(\frac{z^2 S^2}{r^2 \bar{Y}^2} \right) \right)}$$

Derivation Of Formula for estimation of 'n' with proportion:

Case-I If we wish to control the margin of error 'd' in the estimation population proportion. With S.R. Sampling having sample proportion p, as we want.

$$P_x(|p - P| > d) = \alpha \text{ where } \alpha \text{ is a small probability}$$

We assume that p is a normally distributed and its variance;

$$V(p) = \frac{(N-n)PQ}{(N-1)n}$$

The eq. that connects 'n' with the desired level of precision is

$$Z = \frac{|p - P|}{S.E.(p)} = \frac{d}{\sqrt{\frac{(N-n)PQ}{(N-1)n}}}$$

Squaring both sides we get

$$Z^2 = \frac{d^2}{\frac{(N-n)PQ}{(N-1)n}}$$

$$Z^2 = \frac{n(N-1)d^2}{(N-n)PQ}$$

$$Z^2(N-n)PQ = nNd^2 - nd^2$$

$$Z^2NPQ - Z^2nPQ = nNd^2 - nd^2$$

$$nNd^2 - nd^2 + Z^2nPG = Z^2NPG$$

$$(Nd^2 - d^2 + Z^2PG)n = Z^2NPG$$

$$n = \frac{Z^2NPG}{Nd^2 - d^2 + Z^2PG}$$

$$n = \frac{Z^2NPG}{Nd^2 - d^2 + Z^2PG}$$

$$n = \frac{Z^2NPG}{Nd^2 - d^2 + Z^2PG}$$

$$Nd^2 \left(1 - \frac{1}{N} + \frac{Z^2PG}{Nd^2} \right)$$

$$n = \frac{Z^2PG}{d^2}$$

$$1 + \frac{Z^2PG}{Nd^2} - \frac{1}{N}$$

$$n = \frac{Z^2PG}{d^2}$$

(proved)

$$1 + \frac{1}{N} (Z^2PG - 1)$$

Ignoring f.p.c

$$n_0 = \frac{Z^2PG}{d^2}$$

$$n_1 = \frac{n_0}{1 + (n_0 - 1)/N} \Rightarrow n_1 < n_0$$

$$n_2 = \frac{n_1}{1 + (n_1 - 1)/N} \Rightarrow n_2 < n_1$$



$$n_h = n \frac{W_h S_h}{\sum W_h S_h}$$

This allocation was introduced by Neyman therefore it is also named as Neyman Allocation. Substituting the value of n_h as given by Neyman in

$$V(\bar{y}_{st}) = \sum_{h=1}^L \frac{W_h^2 S_h^2}{n_h} = \frac{1}{N} \sum_{h=1}^L W_h^2 S_h^2$$

$$V(\bar{y}_{st})_{\text{Ney}} = \sum_{h=1}^L \frac{W_h^2 S_h^2}{n \frac{W_h S_h}{\sum W_h S_h}} = \frac{1}{N} \sum_{h=1}^L W_h S_h$$

$$V(\bar{y}_{st})_{\text{Ney}} = \frac{1}{n} \left(\sum_{h=1}^L W_h S_h \right)^2 - \frac{1}{N} \sum_{h=1}^L W_h S_h$$

An estimated variance of $V(\bar{y}_{st})_{\text{Ney}}$

$$V(\bar{y}_{st})_{\text{Ney}} = \frac{1}{n} \left(\sum_{h=1}^L W_h S_h \right)^2 - \frac{1}{N} \sum_{h=1}^L W_h S_h$$

Optimum allocation means the variance is minimum (which is best) that's why we named it optimum allocation.

Definition of optimum Allocation:

This method of allocation was proposed by J. Neyman and it consists of finding n_h which minimizes the variance of the stratified sample mean for a fixed total sample size n , assume the cost of surveying the units i.e., c_i to be the same in all strata. The stratum sample size n_h is given by the relation

$$n_h = n \frac{N_h S_h}{\sum N_h S_h} \Rightarrow n_h = n \frac{\frac{N_h S_h}{N}}{\frac{\sum N_h S_h}{N}} \div \text{Numerator and Denominator}$$

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$$V(\bar{y}_{st})_{\text{Ney}} = \frac{1}{n} \left(\sum_{h=1}^L W_h S_h \right)^2 - \frac{1}{N} \sum_{h=1}^L W_h S_h$$

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$$n_h = n \frac{N_h S_h}{\sum N_h S_h} \Rightarrow n_h = n \frac{\frac{N_h S_h}{N}}{\frac{\sum N_h S_h}{N}} \div \text{Numerator and Denominator}$$

$$n_h = n \frac{W_h S_h}{\sum W_h S_h}$$

This allocation also give minimum variance. When cost is not ^{involved} given then minimum variance is obtained by Neyman allocation while cost is involved then minimum variance is obtained by proportional allocation.

→ If C_h is the same for all strata, allocation reduces to Neyman allocation. Thus Neyman allocation is a special case of optimum allocation.

Definition of Optimum Allocation:

The allocation is called optimum when the total sample size 'n' is allocated among the different strata in such a way that for a given cost of selecting the sample, the variance of the estimated mean $\bar{y}_{st}$ i.e. $V(\bar{y}_{st})$ is minimized. The stratum sample size n_h for this method of allocation is

$$n_h = n \frac{N_h S_h / \sqrt{C_h}}{\sum N_h S_h / \sqrt{C_h}}$$

Different Allocations can also be asked in question e.g. $n_h \propto Y_h$

The allocation in which n_h is proportional to the total of h^{th} stratum, we find variance of estimated mean $\bar{y}_{st}$ i.e. $V(\bar{y}_{st})$ for this allocation.

So, $n_h \propto Y_h$

$$n_h = n \frac{Y_h}{\sum Y_h}$$

Substituting this value of n_h given by this allocation in following eqn,

$$V(\bar{y}_{st}) = \sum \frac{W_h S_h^2}{n_h} = \frac{1}{N} \sum W_h S_h^2$$

$$V(\bar{y}_{st}) = \sum \frac{W_h S_h^2}{n_h \frac{Y_h}{\sum Y_h}} = \frac{1}{N} \sum W_h S_h^2$$

$$V(\bar{y}_{st}) = \frac{1}{n} \left(\sum W_h^2 S_h^2 / Y_h \right) (\sum Y_h) = \frac{1}{N} \sum W_h S_h^2$$

Similarly we can find variance for any other allocation asked in question.

Note 1: Sampling always revolve around estimation of parameters, variance of estimators and estimators of estimated variance.

Note 2: We always replace S_h^2 by its estimator s_h^2 because computation of S_h^2 requires the knowledge of each observation in the population which is very difficult task. As formula of $S_h^2 = \sum_{i=1}^{N_h} (y_{hi} - \bar{y}_h)^2 / N_h - 1$ & involves y_{hi} which mean we

must have information for each y_{hi} . So that we replace S_h^2 by s_h^2 because s_h^2 easy to compute. Because s_h^2 is the variance of h^{th} stratum which contains small number of sampling unit and it is easy to obtain knowledge about these small number of sampling unit e.g. if stratum consist 5 sampling unit then we can easily get information/knowledge about 5 sampling unit.